CTTF: Cross-Temporal Tri-Difference Fusion With Mamba for Remote Sensing Change Detection

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***Abstract*—Mamba has recently been widely applied to remote sensing change detection (CD) tasks due to its ability to efficiently represent long-range dependencies with nearly linear computational complexity. However, existing Mamba-based CD methods focus primarily on global modeling during the decoding stage, where insufficient attention to fine-grained details causes blurred change boundaries. In addition, most existing methods rely solely on a single type of differential feature (e.g., signed difference or absolute difference), ignoring the complementary information among different types of differential features, which makes them prone to missed or false detections under registration errors. To address these issues, this article proposes a decoder-side improved framework named cross-temporal tri-difference fusion (CTTF). It consists of a global–local–point (GLP) Block and a signed–absolute–sum fusion (SASF) Module, enabling accurate boundary feature detection, thereby achieving more consistent and reliable change discrimination. Specifically, the GLP Block decouples and models boundary details, global semantics, and information anchors, achieving a dynamic balance between local enhancement and semantic fidelity, thereby significantly improving the boundary feature perception. The SASF constructs a tri-differential mechanism, where the sum difference serves as a stability prior to perform multiscale calibration between the signed difference and the absolute difference. This design compensates for the loss of fine structural and consistency information caused by relying on a single differential domain, effectively reducing both missed and false detections. The proposed method achieves state-of-the-art (SOTA) performance on the WHU-CD, LEVIR-CD, SYSU-CD, and CLCD datasets. Ablation experiments further validate the effectiveness of the proposed SASF Module and GLP Block.**

***Index Terms*—Binary change detection, change detection (CD), cross-temporal interaction, Mamba, state-space model (SSM).**

I. Introduction

CHANGE detection (CD) in remote sensing aims to locate and quantify genuine change regions between coregistered bi-temporal images, serving crucial applications, such as urban renewal monitoring [1], emergency assessment [2], [3], [4], and infrastructure inspection [5]. The task emphasizes robustly revealing scene discrepancies under the premise of accurate temporal registration. In recent years, deep learning has greatly advanced this field. Convolutional neural networks (CNNs) [6], relying on localized convolutions, excel at capturing edges and textures due to their structural simplicity and ease of stacking. However, their fixed receptive fields and hierarchical propagation mechanisms limit their ability to encompass long-range contextual dependencies. In contrast, the Transformer [7] employs self-attention to explicitly aggregate global semantics, mitigating semantic fragmentation induced by long-range dependencies; yet, its quadratic computational growth with respect to image resolution renders both training and inference computationally expensive. Distinct from the above paradigms, Mamba [8], originating from state-space models (SSMs), models long-range dependencies through selective scanning with near-linear computational complexity. This design achieves a more balanced tradeoff between efficiency and global representation, providing a new pathway for CD to maintain global coherence while ensuring computational tractability.

Benefiting from the architectural strengths of Mamba, both Vision Mamba [9] and VMamba [10] extend its application from sequential modeling to visual understanding tasks, introducing refined mechanisms for unidirectional selective scanning. Owing to its efficiency in capturing long-range contextual dependencies, the Mamba architecture aligns exceptionally well with remote sensing imagery. Recent studies, such as ChangeMamba [11] and RS-Mamba [12], have already explored its potential in this domain.

Despite recent progress, remote sensing image CD still faces several fundamental challenges. First, within the changed regions, there often exist extremely small subregions. This makes it difficult for models to accurately capture these subtle variations and often results in missed detections. Second, complex backgrounds and pseudochanges, such as illumination variations, seasonal differences, and slight misregistration, introduce significant noise. These factors make it challenging to distinguish real semantic changes from irrelevant variations, thereby increasing false alarms. Third, change patterns exhibit strong multiscale characteristics, requiring models to simultaneously capture fine-grained local details and long-range contextual dependencies. However, balancing local sensitivity and global consistency remains difficult for existing architectures.

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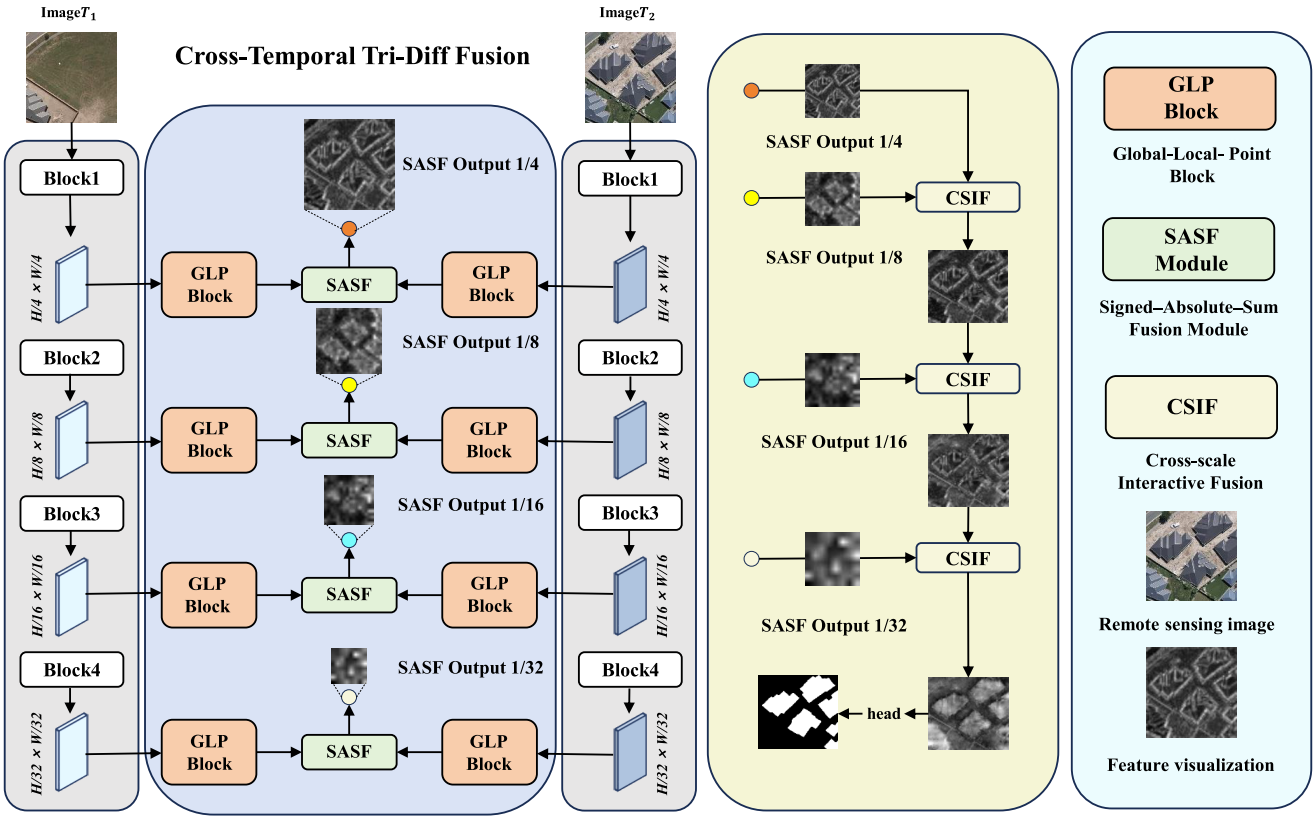

Fig. 1. CTTF architecture takes a pair of coregistered remote sensing images as input and feeds them into a Siamese backbone network with shared weights, producing two feature maps at each stage. After being processed by the GLP Block, these feature maps are passed into the SASF Module to generate enhanced representations, which are then progressively aggregated through the CSIF [26] module—a cross-scale interactive fusion module.

Existing Mamba-based CD methods can generally be grouped into two categories: one is replacing conventional subnetworks with Vision Mamba modules and training the model from scratch; the other is directly transferring pretrained Vision Mamba modules for fine-tuning on CD datasets.

However, when applied to CD tasks, existing Mamba-based methods still face two major bottlenecks. First, existing Mamba-based approaches tend to emphasize global modeling during the decoding stage, leading to insufficient extraction and utilization of local details—particularly high-frequency textures and small-scale variations around boundary regions. For example, in RS-Mamba, the decoder employs simple concatenation and upsampling operations, with most performance gains arising from global contextual modeling and holistic feature aggregation in the encoder. As a result, the decoder's capability for boundary refinement and high-frequency detail enhancement remains limited. Second, although various strategies for constructing differential features have been proposed, most practical systems still rely on a single type of difference (e.g., signed or absolute), which prevents full exploitation of the complementary information among diverse differential domains. This often results in missed detections of fine-grained boundaries and weak targets. For instance, CDMamba [13] explicitly extracts absolute-difference features at multiple scales and feeds them into the decoder; however, such a single-type difference provides limited robustness against registration errors, illumination or viewpoint variations, and nontarget semantic disturbances. Moreover, several lightweight methods [14] also adopt the absolute difference either as a third input or as the sole differential cue to guide feature learning, further confirming both the prevalence and the limitation of single-path differencing in practice.

To address the above challenges, particularly the insufficient utilization of local details and the limitations of single-path differencing, this article proposes a decoder-enhanced framework named cross-temporal tri-difference fusion (CTTF), as illustrated in Fig. 1.

Compared with existing approaches, the proposed method differs in three key aspects. First, instead of relying solely on global modeling, CTTF explicitly enhances local feature sensitivity in the decoding stage through multireceptive-field design and pointwise attention, improving boundary and fine-detail representation. Second, unlike conventional methods that adopt a single difference strategy, we introduce a tri-difference fusion mechanism to fully exploit complementary information across multiple differential domains. Third, the proposed design achieves a more effective balance between global consistency and local discriminability, while maintaining computational efficiency.

First, to mitigate the overemphasis on global modeling and the lack of local sensitivity observed in existing Mamba-based decoders, CTTF introduces a global–local–point (GLP) Block into the decoding stage to perform multireceptive-field enhancement and point attention (PA) on bi-temporal decoding

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features. Specifically, the multireceptive-field enhancement is jointly composed of Mamba, wavelet transform, and multiscale convolution: Mamba efficiently models long-range dependencies through selective scanning with near-linear computational complexity; the wavelet transform provides multiresolution frequency-domain decomposition, highlighting change details across scales; multiscale convolution complements these by reinforcing texture and boundary information within local neighborhoods. In parallel, the PA mechanism generates pointwise weighting maps within feature channels to emphasize minute objects and subtle boundary perturbations, which are preserved through residual aggregation. The synergy between these two components ensures both global consistency and local discriminability, enhancing the network's ability to capture boundary and texture details. Consequently, CTTF effectively reduces missed and false detections under mild registration errors, thereby lowering the false alarm rate and improving overall robustness.

Second, to overcome the limitations of single-path differencing and fully exploit cross-temporal complementary information, we propose the signed–absolute–sum fusion (SASF) Module. On top of the decoder features, it explicitly constructs three differential pathways—signed difference, absolute difference, and sum difference—each equipped with multiscale feature extraction units to ensure sufficient representation. Through explicit cross-channel differential modeling and complementary information aggregation, the SASF Module enhances cross-temporal interaction and fine-grained boundary representation.

In summary, the CTTF maintains global coherence while substantially improving the decoder's sensitivity to local change regions and the completeness of differential modeling. This framework provides a more refined and efficient decoding optimization scheme for remote sensing CD. The main contributions of this work are summarized as follows.

1) *GLP Block:* We propose a GLP Block that decouples and models boundary details, global semantics, and information anchors, achieving a dynamic balance between local enhancement and semantic fidelity. This design substantially improves the detection precision of the changing area's boundary feature detection.
2) *SASF Module:* We introduce an SASF Module that simultaneously constructs three types of differencing—signed, absolute, and sum—on top of the decoder features. By leveraging multiscale cues from these three differential maps, the SASF Module performs explicit differential modeling and effectively aggregates complementary information, alleviating the representational bias caused by single-difference pathways and thereby reducing false and missed detections.
3) *CTTF Framework:* Building upon the integration of the GLP Block and SASF Module, we construct the CTTF framework, which achieves significant performance gains. Under identical training and evaluation settings as the baseline, CTTF delivers superior results, validating its overall advantages in fine-grained representation and practical applicability.

The structure of this article is arranged as follows. Section II reviews related studies. Section III elaborates on the proposed approach. Section IV presents the experimental results. Section V concludes this article.

II. Related Work

A. Deep-Learning-Based CD Methods

Early remote sensing CD relied primarily on CNNs. Representative works, such as FC-Siam-Diff, FC-Siam-Conc [15], and SNUNet [16], adopt Siamese architectures to extract features from bi-temporal images and use convolutions in the encoder to capture local details, such as edges and textures. These methods are simple, easy to train, and have shown strong efficiency in early tasks. However, due to the locality of convolutional receptive fields, they struggle to model long-range dependencies, often leading to global semantic inconsistency or missed changes across regions in complex, large-scale scenes. Beyond CD itself, recent studies in related visual and remote sensing tasks have also explored structural designs for improving feature representation, localization, and robustness. For example, instance-level mask modeling, text detection, and infrared small-target detection have been investigated from different perspectives, providing useful references for designing task-oriented feature extraction and refinement strategies [17], [18], [19], [20], [21]. These studies suggest that effective feature refinement remains important when dealing with complex boundaries, small targets, and diverse background interference.

To overcome this bottleneck, Transformers have been introduced into CD. BIT [22] first treats the two temporal images as semantic tokens and models cross-temporal dependencies via global self-attention; ChangeFormer [23] integrates multihead self-attention into an encoder–decoder architecture, markedly improving cross-temporal semantic consistency; subsequent methods, such as SwinSUNet [24], combine local window attention with global dependence modeling to balance performance and efficiency. Meanwhile, recent studies in related remote sensing tasks further demonstrate the effectiveness of Transformer-based architectures in modeling complex spectral–spatial relationships. For instance, CasFormer [25] adopts a cascaded Transformer design with spectral-aware self-attention and spatial cross-attention to enhance fusion-aware hyperspectral reconstruction, highlighting the capability of hierarchical attention mechanisms in capturing multidimensional dependencies. Despite their advantages in global modeling over CNNs, Transformer complexity grows quadratically with input resolution, incurring heavy compute and memory costs for high-resolution remote sensing imagery.

Recently, the Mamba architecture—based on SSMs—has offered a new direction. Through selective scanning, Mamba achieves long-range dependence modeling with near-linear complexity; its vision variants, Vision Mamba and VMamba, have demonstrated strong global modeling and efficiency in vision tasks. Meanwhile, recent studies further explore the practical deployment of such architectures; for example, Mamba-X [27] introduces hardware-aware optimization strategies to alleviate the sequential dependence in selective scanning, improving parallelism and efficiency in edge

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computing scenarios. Inspired by this, methods such as ChangeMamba, RS-Mamba, CDMamba, and the more recent SPMNet [28] have introduced Mamba into remote sensing CD and achieved leading performance on multiple benchmarks. For example, CDMamba proposes the SRCM module, which combines global features with local convolutional cues, and designs the AGLGF module to enhance cross-temporal interaction, effectively compensating for detail cues that pure scanning may overlook; SPMNet introduces a pyramid-style multiscale mechanism within a Siamese framework to better reconcile wide-area changes and local details in ultrahigh-resolution images. These works indicate that Mamba strikes a good balance between accuracy and efficiency. Nevertheless, many existing methods still rely on simple concatenation or upsampling in the decoder, limiting sensitivity to fine-grained features and risking missed detections for small objects or boundary changes.

In addition, recent advances in hyperspectral imaging [29] provide a broader perspective on remote sensing analysis, emphasizing the importance of jointly modeling high-dimensional spectral and spatial information. Comprehensive studies have shown that effective representation learning in such scenarios often requires integrating multiscale and multidomain features, which further underscores the necessity of balancing global context and local detail modeling in complex environments. Meanwhile, recent remote sensing studies have also explored local decomposition, structural memory, and hierarchical adaptive aggregation strategies for change-related representation learning [30], [31]. These works further indicate that combining local detail preservation with higher-level structural modeling is a promising direction for complex remote sensing interpretation.

To address these issues, our proposed GLP Block introduces multireceptive-field enhancement and PA on the decoder side: Mamba handles long-range dependence modeling; wavelet-based multiscale decomposition emphasizes change details across frequency bands; convolutional units further capture neighborhood textures and boundaries; and PA strengthens responses to tiny targets and subtle perturbations. This design remedies decoder-side limitations of existing Mamba-based methods and unifies global consistency with local sensitivity without appreciably increasing computational cost.

To simultaneously capture long-range contextual dependencies and fine-grained spatial cues in CD tasks, we propose a GLP Block, which extends the Mamba architecture with a multibranch decomposition strategy. Unlike prior approaches that enhance global modeling solely by modifying scanning mechanisms, GLP Block decomposes the input channels into four complementary subspaces and processes them with heterogeneous operators, each tailored to a specific aspect of feature modeling. This design explicitly enforces GLP–identity complementarity, leading to more robust representation learning.

B. Applications of Differential Features in CD

Differential features explicitly transform discrepancies between two temporal images at the same location into a difference map, indicating where changes occur and their magnitude. As the most direct cue for CD, they have long been used to reveal changed regions. Traditional approaches typically perform pixelwise differencing followed by thresholding to produce change maps—physically interpretable yet sensitive to threshold settings and prone to noise, misregistration, and illumination variations. With deep learning, differencing has gradually moved into feature space; a typical example is FC-Siam-Diff, which constructs differential inputs using absolute differences. Building on this, recent work has systematized the modeling of "how to construct, amplify, and filter differences." One line centers on multiscale differential enhancement: SMD-Net [32] progressively generates and reinforces multiscale difference maps to focus on changed regions, markedly improving the separability of boundaries and small changes. Another line targets hyperspectral/multispectral imagery by designing tailored bidirectional difference representations; for instance, MSDFFN [33] learns "difference components" across scales via its BDFR module and uses attention-based fusion to boost discrimination of subtle changes, exemplifying fine-grained modeling in high-dimensional spectral–spatial domains. While these methods improve focus on changed areas, they are often limited by relying on a single differencing pathway—typically the absolute difference—making it difficult to fully express differential characteristics in complex scenes. To mitigate this limitation, we propose the SASF Module, which explicitly constructs and fuses three types of differential features—signed difference, absolute difference, and sum difference—on the decoder side. Through multiscale feature extraction units, the SASF Module finely models these three differences in both spatial and semantic dimensions and aggregates complementary information across channels. This design explicitly strengthens cross-temporal interaction capacity and substantially alleviates the representational bias inherent to single-path differencing, yielding greater robustness and sensitivity to weak changes and boundary details.

III. Approach

A. Overview

The overall architecture adopts a Siamese encoder–decoder framework, as illustrated in Fig. 1, in which bi-temporal inputs are first processed independently through a shared SeaFormer-L encoder [34] to extract multiscale feature representations. In general, the encoder mainly serves as a high-efficiency feature extractor, providing context-aware and structurally rich embeddings.

At the decoder side, the central computational block is the CTTF, which is responsible for explicitly modeling the discrepancies between bi-temporal features. Each temporal feature map is first enhanced by the GLP Block, which employs a four-branch structure: global Mamba-wavelet-enhanced context, multikernel local convolutions, pointwise attention, and identity residual to effectively combine long-range semantics with local details. Based on these enhanced representations, CTTF then constructs a tri-difference mechanism through a lightweight SASF Module, capturing signed, absolute, and sum differences to enhance boundary sensitivity

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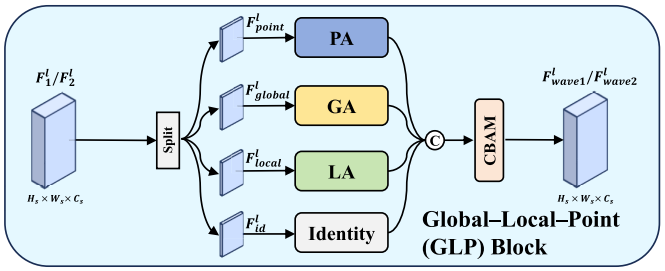

Fig. 2. Schematic of the proposed GLP Block. The input feature is split into four complementary branches: PA, global branch (GA), local branch (LA), and an identity pathway. Their outputs are concatenated ("C") and refined through a CBAM module to produce enhanced multigranular representations.

and local discrimination. The three streams are concatenated, recalibrated by an attention mechanism, and finally compressed by a 1×1 convolution into a compact change embedding.

On top of this, the decoder further integrates cross-scale features to generate the final prediction, ensuring spatial consistency across resolutions.

In summary, the proposed model leverages the SeaFormer-L encoder for feature extraction, while the novel CTTF in the decoder achieves both intratemporal enhancement and intertemporal discrepancy modeling. Complemented by cross-scale fusion, this hierarchical design effectively combines rich temporal semantics with precise boundary information, thereby improving the discriminability and localization accuracy of CD.

B. GLP Block

To simultaneously capture long-range contextual dependencies and fine-grained spatial cues in CD tasks, we propose a GLP Block (as shown in Fig. 2), which extends the Mamba architecture with a multibranch decomposition strategy. Unlike prior approaches that enhance global modeling solely by modifying scanning mechanisms, GLP Block decomposes the input channels into four complementary subspaces and processes them with heterogeneous operators, each tailored to a specific aspect of feature modeling. This design explicitly enforces GLP–identity complementarity, leading to more robust representation learning.

Formally, given an input feature map

$$F \in \mathbb{R}^{H\times W\times C} \tag{1}$$

the channel dimension is split into four groups

$$F = [F_{\mathrm{G}} \parallel F_{\mathrm{L}} \parallel F_{\mathrm{P}} \parallel F_{\mathrm{Id}}] \tag{2}$$

where F_{G}, F_{L}, F_{P}, and F_{Id} denote the global, local, PA, and identity partitions, respectively. The operations on each branch are described as follows.

1) Global Branch (F_{G}): To reconcile long-range dependence modeling with high-frequency detail under a constrained computational budget, the GLP allocates a channel subspace of proportion ξ, denoted $F_G \in \mathbb{R}^{h\times w\times \xi c}$, and executes two complementary pathways *in parallel* on this subspace: one performs global sequence modeling via a selective state-space operator (Mamba), and the other enhances fine-scale structure in the frequency domain via a Haar wavelet transform. The two streams operate concurrently on the same input without serial interdependence and are ultimately aggregated by elementwise sum, thereby coupling semantic coherence with textural fidelity.

Global sequence modeling (Bi-scan SSM)

$$x^{\mathrm{in}}_{m1} = \mathrm{Linear}(F_G)[:, :\xi c] \tag{3}$$

$$x^{\mathrm{in}}_{m2} = \mathrm{Linear}(F_G)[:, \xi c :] \tag{4}$$

$$x_{m1} = \mathrm{SSM}\left(\sigma(\mathrm{Conv}(x^{\mathrm{in}}_{m1}))\right) \tag{5}$$

$$x_{m2} = \sigma\left(x^{\mathrm{in}}_{m2}\right) \tag{6}$$

$$x^{\mathrm{out}}_{m} = \mathrm{Linear}\left(x_{m1} \otimes x_{m2}\right) \tag{7}$$

where σ denotes the gating nonlinearity, $\otimes$ is the elementwise product, and the state-space module adopts a bidirectional scanning scheme to improve stability and global coverage.

Frequency-domain detail enhancement

$$x^{\mathrm{in}}_{wt} = \mathrm{WT}\left(F_G; [f_{LL}, f_{LH}, f_{HL}, f_{HH}]\right) \tag{8}$$

$$x^{\mathrm{out}}_{w} = \mathrm{IWT}\left(\mathrm{Conv}\left(x^{\mathrm{in}}_{wt}\right); [f_{LL}, f_{LH}, f_{HL}, f_{HH}]\right) \tag{9}$$

with the Haar filters given by

$$f_{LL} = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}, \quad f_{LH} = \frac{1}{2}\begin{bmatrix}1 & -1\\ 1 & -1\end{bmatrix} \tag{10}$$

$$f_{HL} = \frac{1}{2}\begin{bmatrix}1 & 1\\ -1 & -1\end{bmatrix}, \quad f_{HH} = \frac{1}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}. \tag{11}$$

Operating in the downsampled subband domain, this branch applies lightweight convolutions to expand the receptive field while accentuating edges and other high-frequency cues.

Parallel fusion and output

$$F^{\mathrm{out}}_G = x^{\mathrm{out}}_m + x^{\mathrm{out}}_w, \quad F^{\mathrm{out}}_G \in \mathbb{R}^{h\times w\times \xi c}. \tag{12}$$

The SSM pathway enforces global contextual consistency, whereas the wavelet pathway restores structural boundaries and textures. The resulting representation achieves high semantic density and faithful detail without introducing additional serial dependencies.

2) Local Branch (F_L): To complement global modeling with fine-scale structures under the same budget, the GLP Block assigns the remaining μc channels to a local pathway $F_L \in \mathbb{R}^{h\times w\times \mu c}$ with $\mu \le 1-\xi$. This branch performs multireceptive-field dynamic depthwise convolutions in parallel and aggregates their responses via kernelwise voting, followed by a scaled residual connection to stabilize training.

Multireceptive dynamic depthwise paths. Split F_L evenly into n parts $\{x^I_{L_j}\}_{j=1}^n$, each of size $\mathbb{R}^{h\times w\times \mu c/n}$. The jth part is processed by a dynamic depthwise operator with kernel $k_j = 2j+1$ and E experts, gated by a softmax

$$\alpha^{(j)} = \mathrm{softmax}\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{GAP}\left(x^I_{L_j}\right)\right)\right) \in \mathbb{R}^E \tag{13}$$

$$y^{\mathrm{dw}}_{L_j} = \sum_{e=1}^{E} \alpha^{(j)}_e \cdot \left(W^{(j,e)}_{k_j} \overset{\mathrm{dw}}{*} x^I_{L_j}\right) \tag{14}$$

$$z_{L_j} = \mathrm{PWConv}_{g=\mu c/n}\left(\mathrm{BN}\left(\sigma\left(\mathrm{BN}\left(y^{\mathrm{dw}}_{L_j}\right)\right)\right)\right) \tag{15}$$

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where σ denotes a lightweight nonlinearity Mish module and $\mathrm{PWConv}_{g=\mu c/n}$ is a per-channel 1×1 projection.

Kernelwise aggregation concatenates the multikernel outputs along the channel dimension, computes a channel-mean "kernel vote," and broadcasts it back to μc channels

$$\tilde{x}_L = \mathrm{Concat}\left([z_{L_1}, \ldots, z_{L_n}],\ \mathrm{dim} = -1\right) \tag{16}$$

$$s_L = \mathrm{Mean}_{\mathrm{ch}}(\tilde{x}_L) \tag{17}$$

$$y_L = \mathrm{Broadcast}_{\mu c}(s_L) \in \mathbb{R}^{h \times w \times \mu c}. \tag{18}$$

After effectively aggregating the multireceptive responses from different kernel paths, the local branch yields a compact representation that captures fine-grained structural cues and boundary information

$$F_L^{\mathrm{out}} = y_L, \quad F_L^{\mathrm{out}} \in \mathbb{R}^{h \times w \times \mu c}. \tag{19}$$

3) Point Branch (F_{P}): To selectively emphasize informative activations, the PA branch processes the input feature $F_P \in \mathbb{R}^{h \times w \times \lambda c}$ with $\lambda \leq 1 - \xi - \mu$ using a pointwise gating mechanism. Specifically, two consecutive 1×1 convolutions are applied, where the first expands the channel dimension by a factor of four and the second restores it to the original dimension. Each convolution is followed by batch normalization and a nonlinear activation function (GELU [35]). A sigmoid gating function is then employed to generate attention weights that modulate the input features via elementwise multiplication, thereby adaptively reweighting the spatial responses

$$F_P^{\mathrm{out}} = F_P \odot \sigma\left(W_2 * \mathrm{GELU}\left(\mathrm{BN}(W_1 * F_P)\right)\right). \tag{20}$$

Here, $W_1 \in \mathbb{R}^{4C \times C \times 1 \times 1}$ and $W_2 \in \mathbb{R}^{C \times 4C \times 1 \times 1}$ denote the learnable pointwise convolution kernels for channel expansion and reduction, respectively. $\mathrm{BN}(\cdot)$ represents batch normalization, $\mathrm{GELU}(\cdot)$ is the Gaussian error linear unit activation, $\sigma(\cdot)$ denotes the sigmoid gating function, and $\odot$ indicates elementwise multiplication. The GELU activation is defined as

$$\mathrm{GELU}(x) = x \cdot \Phi(x) = x \cdot \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{x}{\sqrt{2}}\right)\right] \tag{21}$$

where $\Phi(x)$ is the cumulative distribution function of the standard Gaussian distribution.

4) Identity Branch (F_{Id}): The remaining channels are kept unaltered as a residual bypass to preserve the original signal distribution. This identity pathway prevents over-distortion during transformation and stabilizes optimization by maintaining gradient flow.

Finally, the outputs of the four branches are concatenated and then passed through a *projection head* (Mish [36] activation, 1×1 pointwise convolution, and a CBAM [37] unit). This step enables cross-branch interaction and adaptive recalibration in both channel and spatial dimensions, yielding the final enhanced representation

$$F' = \mathrm{Proj}\ (\mathrm{Concat}(F_{\mathrm{G}}, F_{\mathrm{L}}, F_{\mathrm{P}}, F_{\mathrm{Id}})) \tag{22}$$

where F_{G}, F_{L}, F_{P}, and F_{Id} are the branch outputs. In particular, F_{G} carries long-range contextual semantics from the wavelet-enhanced state-space operator, F_{L} encodes multiscale structural details refined by heterogeneous convolutions, F_{P} adaptively highlights discriminative activations through pointwise gating, and F_{Id} preserves the original identity features for stable optimization.

By explicitly decomposing feature channels into G/L/P/Id pathways, the GLP Block balances contextual semantics and structural fidelity: the wavelet-enhanced global branch models long-range dependencies, the local branch refines boundaries, the PA branch highlights discriminative cues, and the identity branch stabilizes learning—forming a compact yet powerful unit for dense prediction tasks, such as remote sensing CD.

C. SASF Module

While the GLP Block focuses on enhancing single-temporal features, CD requires explicit modeling of bi-temporal discrepancies. To this end, we propose the SASF Module (as shown in Fig. 3), which decomposes differential features into three complementary streams and employs lightweight local refinement to highlight subtle boundaries. Unlike prior approaches that rely solely on a single subtraction or concatenation strategy, the SASF Module disentangles directional, magnitude, and contextual cues, and selectively reemphasizes local variations that are most critical for accurate boundary delineation.

Given two enhanced features from bi-temporal images, $F_A, F_B \in \mathbb{R}^{C \times H \times W}$, we compute three complementary differential maps

$$F_{sign} = F_A - F_B \tag{23}$$

$$F_{abs} = |F_A - F_B| \tag{24}$$

$$F_{sum} = F_A + F_B \tag{25}$$

where F_{sign} captures signed changes (direction-sensitive), F_{abs} encodes absolute variations (magnitude-only), and F_{sum} provides contextual consistency (shared structures and background priors). These streams together offer a richer representation space for discriminative change modeling.

Structurally, this block inherits the multibranch decomposition idea of the GLP Block, but with a constrained configuration: the global branch is disabled to avoid redundant long-range modeling, while the local branch is preserved with a narrow allocation, and an identity branch is retained to pass features unchanged. This ensures that the refinement focuses on boundary sharpening rather than global semantics. Compared to the local operators in the GLP Block, which emphasize general spatial structures within a single temporal feature, the local operators in the SASF Module are deliberately specialized for intertemporal differential cues, especially at object boundaries.

After refinement, the three outputs are concatenated

$$F_{cat} = \mathrm{Concat}\left(F'_{sign}, F'_{abs}, F'_{sum}\right) \tag{26}$$

$$F_{cat} \in \mathbb{R}^{3C \times H \times W} \tag{27}$$

where F'_{sign}, F'_{abs}, and F'_{sum} denote the refined signed-difference, absolute-difference, and sum features, respectively. The concatenated feature is then recalibrated by a CBAM attention unit and mapped through a 1×1 convolution to obtain the final differential embedding

$$F_{out} = \mathrm{Proj}\,(F_{cat}) \tag{28}$$

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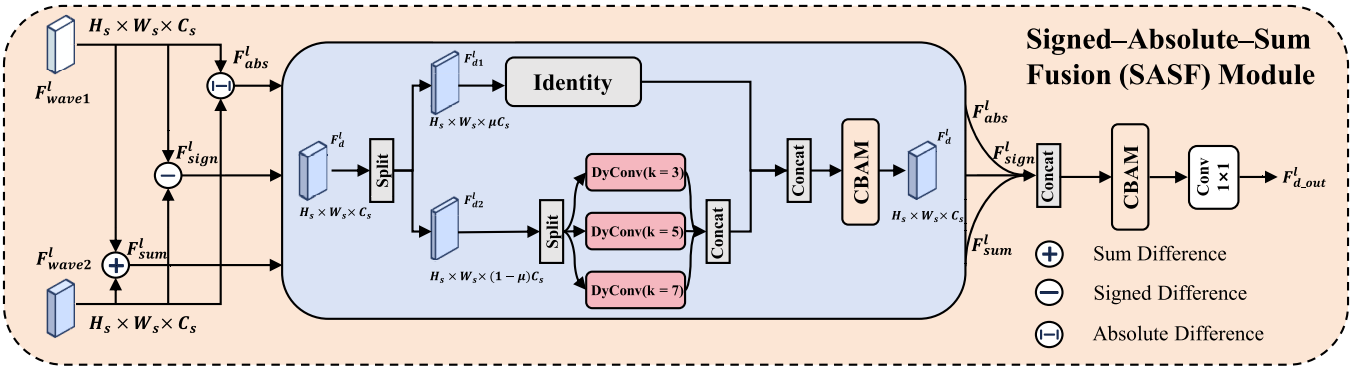

Fig. 3. Schematic of the SASF Module. Bi-temporal features are first transformed into signed difference, absolute difference, and sum components, which are then processed by a channel-split pathway composed of identity mapping and multikernel dynamic convolutions. The fused representation is refined through CBAM and a 1×1 convolution to generate the final difference-enhanced feature F_{d_out}.

where Proj(·) is the same projection operator defined in the GLP Block.

In summary, the SASF Module constructs a tri-stream discrepancy representation (signed, absolute, and sum), applies local-boundary-focused refinement, and performs attention-guided fusion. By carefully distinguishing its role from the GLP Block, the SASF Module complements global–local modeling with targeted intertemporal edge enhancement, yielding structurally aligned and semantically discriminative differential features for CD.

IV. Experimental Results and Analysis

A. Data Description

Extensive experiments are conducted on three representative CD benchmarks. Unless otherwise noted, we follow the official protocols of each dataset, adopt nonoverlapping 256×256 cropping, and use binary change masks without additional filtering.

LEVIR-CD [38] is a building CD dataset comprising 637 dual-temporal image pairs of size 1024×1024 with a ground sampling distance (GSD) of 0.5 m. Following the default setting, each pair is partitioned into nonoverlapping 256×256 patches and randomly split into 7120/1024/2048 images for training/validation/testing, respectively.

WHU-CD [39] targets building changes and contains one dual-temporal aerial mosaic of size $32\,507 \times 15\,354$ at 0.2 m GSD. In line with prior work, we crop the mosaic into 256×256 patches and randomly divide them into 6096/762/762 samples for training/validation/testing.

CLCD [40] focuses on cropland changes and consists of 600 dual-temporal pairs of size 512×512, with spatial resolution ranging from 0.5 to 2 m. We adopt the common protocol and randomly split the pairs into 360/120/120 for training/validation/testing.

SYSU-CD [41] is a large-scale aerial image CD dataset collected over Hong Kong, containing 20 000 dual-temporal image pairs of size 256×256 with a spatial resolution of 0.5 m. The dataset covers diverse change types, including newly built buildings, suburban expansion, vegetation variation, road extension, and coastal construction. Following the official split, we use 12 000/4000/4000 image pairs for training/validation/testing, respectively.

All subsequent experimental settings (preprocessing, model input size, and evaluation pipeline) are kept consistent across datasets to enable fair comparison.

B. Training Details

We implement the model in Python using PyTorch and train on a workstation equipped with an NVIDIA RTX 4090 GPU. Optimization proceeds via stochastic gradient descent with an initial learning rate of 0.05, momentum of 0.9, and weight decay of 5×10^{-4}. The learning rate follows a step schedule with step size 50 and decay factor $\gamma = 0.1$ over 300 epochs. Mini-batch sizes are set to 8 for LEVIR-CD and WHU-CD, and 4 for CLCD and SYSU-CD. To bolster out-of-sample performance, we apply standard data augmentation, chiefly random cropping. The objective function combines cross-entropy and Dice losses

$$L_{\text{total}} = \lambda_1 L_{\text{ce}} + \lambda_2 L_{\text{dice}} \tag{29}$$

$$L_{\text{ce}} = -\frac{1}{N} \sum_{i=1}^{N} y_i \log \hat{y}_i \tag{30}$$

$$L_{\text{dice}} = 1 - \frac{2 \sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i}. \tag{31}$$

1) Evaluation Metrics: Model quality is assessed on pixelwise binary masks with a default decision threshold of 0.5. For large-format imagery, we employ sliding-window inference and average predictions across overlaps prior to scoring. *Precision* quantifies the share of predicted-change pixels that are truly changed, reflecting the level of false positives; *Recall* measures the share of truly changed pixels that are recovered, reflecting the level of false negatives; the *F1* score summarizes precision (Pre.) and recall (Rec.) via their harmonic mean; *IoU* captures spatial agreement between predicted and true change regions and is sensitive to boundary fidelity; and *OA* reports overall pixel-level correctness, though it influenced by class imbalance

$$\text{Pre.} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{32}$$

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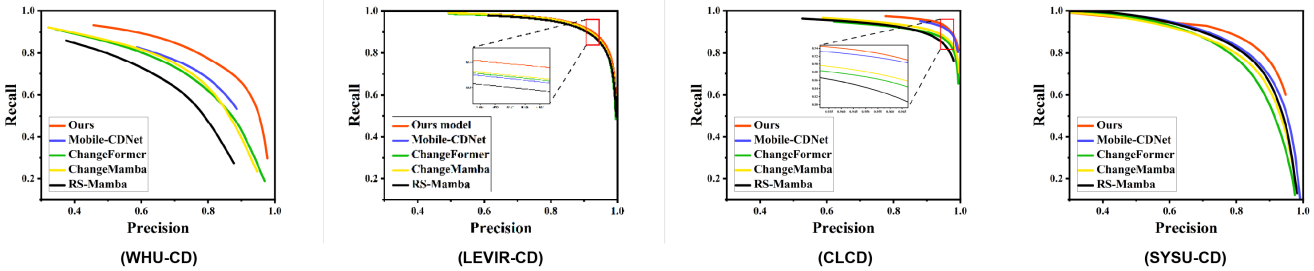

Fig. 4. Comparison of PR curves across various approaches on WHU-CD, LEVIR-CD, SYSU-CD, and CLCD datasets. The horizontal axis denotes precision, while the vertical axis represents recall. Curves that approach the top-right corner indicate a superior balance between precision and recall, reflecting more effective CD performance.

TABLE I

CHANGE RESULTS ON WHU-CD AND CLCD. THE BEST AND SECOND-BEST OF EACH METRIC COLUMN ARE MARKED IN **RED** AND **BLUE**, RESPECTIVELY

Methods	**Params** (M)	**FLOPs** (G)	**WHU-CD** Pre./Rec./F1/IoU/OA	**CLCD** Pre./Rec./F1/IoU/OA
FC-EF [15]	1.10	1.55	91.74 / 87.64 / 89.64 / 81.23 / 99.07	68.38 / 68.48 / 68.43 / 52.01 / 95.30
FC-Siam-Conc [15]	1.35	4.25	82.34 / 88.46 / 85.29 / 74.35 / 98.60	66.70 / 71.05 / 68.81 / 52.45 / 95.21
FC-Siam-Diff [15]	1.55	4.86	84.34 / 87.97 / 86.12 / 75.62 / 98.70	67.24 / 69.61 / 68.40 / 51.98 / 95.22
BIT [22]	11.89	8.71	91.58 / 92.55 / 92.06 / 85.29 / 99.37	74.47 / 73.23 / 73.85 / 58.54 / 96.14
SNUNet [16]	12.03	46.70	83.34 / 90.88 / 86.95 / 76.91 / 98.92	64.74 / 65.88 / 65.30 / 48.48 / 94.79
ChangeFormer [23]	41.03	202.86	94.97 / 86.87 / 90.74 / 83.06 / 99.19	74.91 / 68.22 / 71.41 / 55.53 / 95.94
PaFormer [42]	16.16	10.86	92.87 / 90.85 / 91.85 / 84.92 / 99.36	71.88 / 62.34 / 66.77 / 50.12 / 95.24
SARASNet [43]	56.89	139.90	79.35 / 87.20 / 83.09 / 71.07 / 98.42	76.24 / 75.35 / 75.79 / 61.02 / 96.42
AFCF3DNet [44]	17.64	31.58	95.59 / 88.98 / 92.17 / 85.48 / 99.31	**80.84** / 70.45 / 75.29 / 60.37 / 96.56
ACABFNet [45]	102.32	23.93	93.12 / 87.76 / 90.36 / 82.42 / 99.26	74.53 / 61.59 / 67.45 / 50.88 / 95.44
RS-Mamba [12]	51.95	22.82	93.63 / 86.13 / 89.72 / 81.36 / 99.09	70.08 / 63.70 / 66.73 / 50.08 / 95.27
ChangeMamba [11]	48.56	38.49	94.78 / 88.35 / 91.45 / 84.25 / 99.24	77.09 / 67.64 / 72.06 / 56.32 / 96.10
LCDNet [46]	2.56	4.45	92.34 / 87.78 / 90.00 / 81.82 / 99.11	73.67 / 76.80 / 75.20 / 60.26 / 96.23
Mobile-CDNet [47]	2.95	2.38	95.34 / 91.85 / 93.56 / 87.90 / 99.42	73.83 / 70.86 / 72.31 / 58.51 / 96.26
DDRL [48]	31.18	42.08	91.93 / 87.46 / 89.64 / 81.22 / 99.08	71.73 / 72.36 / 72.05 / 56.31 / 95.82
ConvFormer [49]	49.32	6.99	91.95 / 88.77 / 90.33 / 82.37 / 99.13	63.94 / 72.16 / 67.80 / 51.29 / 94.90
CVNet [50]	106.82	86.70	91.72 / 89.63 / 90.66 / 82.92 / 99.15	74.76 / 70.05 / 72.33 / 56.65 / 96.01
FarmCD [51]	47.96	15.66	**95.87** / 92.11 / 93.96 / 88.60 / 99.46	79.15 / **77.54** / 78.34 / 64.39 / 96.81
Ours	16.21	3.83	95.62 / **92.65** / **94.11** / **88.88** / **99.47**	79.88 / 77.43 / **78.64** / **64.79** / **96.87**

$$\text{Rec.} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{33}$$

$$F1 = \frac{2}{\text{Rec.}^{-1} + \text{Pre.}^{-1}} \tag{34}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{35}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{36}$$

where TP and TN denote the numbers of correctly identified positives and negatives, and FP and FN denote the numbers of negatives misclassified as positive and positives misclassified as negative, respectively.

C. Performance Comparison

To comprehensively evaluate the effectiveness of the proposed CTTF, extensive comparison experiments were conducted on four publicly available benchmarks: WHU-CD, LEVIR-CD, SYSU-CD, and CLCD. The comparisons cover representative methods across four paradigms, including CNN-based (FC-EF [15], FC-Siam-Conc [15], FC-Siam-Diff [15], AFCF3DNet [44], SNUNet [16], LCDNet [46], Mobile-CDNet [47], FarmCD [51], and CVNet [50]), Transformer-based (BiT [22], ChangeFormer [23], SARASNet [43], PaFormer [42], ACABFNet [45], DDRL [48], and ConvFormer [49]), and Mamba-based (RS-Mamba [12] and ChangeMamba [11]).

All methods were trained and evaluated under identical preprocessing, training, and testing protocols to ensure fair comparison. Evaluation metrics include precision (Pre.), recall (Rec.), $F1$ ($F1$), intersection-over-union (IoU), overall accuracy (OA), and computational indicators (parameters and FLOPs), as summarized in Fig. 4 and Tables I and II.

1) Quantitative Results: Tables I and II summarize the overall performance on WHU-CD, LEVIR-CD, SYSU-CD, and CLCD. The proposed CTTF consistently outperforms CNN-, Transformer-, and Mamba-based methods on key metrics ($F1$ and IoU) while remaining lightweight (16.21M, 3.83G

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TABLE II

CHANGE RESULTS ON LEVIR-CD AND SYSU-CD. THE BEST AND SECOND-BEST OF EACH METRIC COLUMN ARE MARKED IN **RED** AND **BLUE**, RESPECTIVELY

Methods	Params (M)	FLOPs (G)	LEVIR-CD Pre./Rec./F1/IoU/OA	SYSU-CD Pre./Rec./F1/IoU/OA
FC-EF [15]	1.10	1.55	91.40 / 88.30 / 89.83 / 81.53 / 99.13	79.85 / 76.40 / 78.09 / 64.05 / 89.89
FC-Siam-Conc [15]	1.35	4.25	90.63 / 90.23 / 90.43 / 82.53 / 99.17	84.68 / 73.30 / 78.58 / 64.72 / 90.58
FC-Siam-Diff [15]	1.55	4.86	90.57 / 86.95 / 88.72 / 79.73 / 99.04	83.34 / 74.57 / 78.71 / 64.90 / 90.49
BIT [22]	11.89	8.71	91.54 / 88.54 / 90.02 / 81.84 / 98.99	81.81 / 76.92 / 79.29 / 65.69 / 90.52
SNUNet [16]	12.03	46.70	89.93 / 87.45 / 88.67 / 79.65 / 98.86	81.19 / 78.71 / 79.93 / 66.57 / 90.68
ChangeFormer [23]	41.03	202.86	92.05 / 89.67 / 90.84 / 83.22 / 99.21	80.44 / 76.76 / 78.56 / 64.69 / 90.12
PaFormer [42]	16.16	10.86	91.76 / 87.59 / 89.63 / 81.21 / 98.97	82.26 / 73.84 / 77.82 / 63.69 / 90.07
SARASNet [43]	56.89	139.90	89.96 / 87.15 / 88.53 / 79.43 / 99.09	87.24 / 69.89 / 77.61 / 63.41 / 90.62
AFCF3DNet [44]	17.64	31.58	91.07 / 89.90 / 90.48 / 82.61 / 99.17	83.68 / 79.46 / 81.51 / 68.79 / 91.50
ACABFNet [45]	102.32	23.93	90.74 / 87.11 / 88.89 / 79.99 / 98.89	81.98 / 80.13 / 81.04 / 68.13 / 91.16
RS-Mamba [12]	51.95	22.82	**92.64** / 88.08 / 90.30 / 82.32 / 99.17	76.89 / 78.04 / 77.46 / 63.21 / 89.29
ChangeMamba [11]	48.56	38.49	91.85 / 89.93 / 90.88 / 83.28 / 99.21	82.75 / 76.44 / 79.47 / 65.93 / 90.69
LCDNet [46]	2.56	4.45	91.09 / 89.69 / 90.38 / 82.45 / 99.17	76.42 / 74.88 / 75.64 / 60.82 / 89.98
Mobile-CDNet [47]	2.95	2.38	91.24 / 90.33 / 90.78 / 83.12 / 99.20	81.88 / 81.68 / 81.78 / 69.18 / 91.42
DDRL [48]	31.18	42.08	92.25 / 87.75 / 89.94 / 81.72 / 99.14	80.40 / 76.62 / 78.47 / 64.56 / 90.08
ConvFormer [49]	49.32	6.99	88.89 / 88.07 / 88.48 / 79.33 / 99.00	81.95 / 80.38 / 81.16 / 68.29 / 91.20
CVNet [50]	106.82	86.70	90.56 / 89.62 / 90.09 / 81.96 / 99.14	78.09 / 80.71 / 79.38 / 65.81 / 90.11
FarmCD [51]	47.96	15.66	92.10 / **90.40** / 91.24 / 83.90 / 99.24	**89.33** / 76.61 / 82.48 / 70.19 / **92.33**
Ours	16.21	3.83	92.30 / 90.30 / **91.29** / **83.98** / **99.25**	83.70 / **81.90** / **82.79** / **70.64** / 91.97

FLOPs). Against CNN models, such as FC-EF, FC-Siam series, SARASNet, and Mobile-CDNet, CTTF shows clear gains across all datasets. On WHU-CD, it reaches the top $F1$ of 94.11% and the highest Recall (92.65%) and OA (99.47%). On LEVIR-CD and CLCD, CTTF also achieves superior $F1$ scores (91.29% and 78.64%), demonstrating stronger stability under multiscale building changes and complex backgrounds.

Compared with Transformer-based methods, including ChangeFormer, PaFormer, BiT, and AFCF3DNet, CTTF maintains leading $F1$ performance. It improves WHU-CD $F1$ by +3.37% over ChangeFormer and +2.26% over PaFormer, and surpasses them on LEVIR-CD and CLCD as well. These results show that CTTF offers better boundary consistency and robustness despite its lower computational cost, outperforming large models, such as ChangeFormer and ACABFNet. Relative to Mamba-based approaches (RS-Mamba and ChangeMamba), CTTF achieves the most balanced overall performance. It yields the highest $F1$ on all datasets, with +0.30% to +11.91% improvements, and consistently superior IoU, indicating better cross-scene generalization and stronger boundary preservation.

On SYSU-CD (as shown in Table II), CTTF also demonstrates strong performance under more challenging change scenarios. It achieves the highest $F1$ (82.79%) and IoU (70.64%), outperforming all competing methods. Compared with recent advanced models, such as FarmCD and Mobile-CDNet, CTTF improves $F1$ by +0.31% and +1.01%, respectively, while maintaining a lower computational cost. Notably, although some methods (e.g., FarmCD and SNUNet) achieve higher precision, their recall is relatively limited, leading to inferior overall $F1$ scores. In contrast, CTTF achieves a better balance between precision (83.70%) and recall (81.90%), indicating more reliable detection of changed regions. These results verify that CTTF is more robust to complex agricultural patterns and class imbalance, and can effectively capture subtle and irregular changes in real-world scenarios.

Fig. 4 presents the PR curves, which further confirm the model's advantage. Across WHU-CD, LEVIR-CD, SYSU-CD, and CLCD, CTTF produces the envelope closest to the top-right region, reflecting an improved tradeoff between precision and recall. It better suppresses false alarms in dense urban regions, remains sensitive to subtle changes, and generalizes well under cross-region variations. Overall, the results demonstrate that combining global context and local structural cues is crucial for RS CD. Through lightweight multiscale modeling, CTTF achieves strong precision–recall balance, fewer errors, and robust performance, showing high potential for practical deployment.

2) Qualitative Results: Representative qualitative results on datasets are presented in Figs. 5–8, where each row corresponds to a specific scene and columns show T_1, T_2, GT, and predictions from different models. Red and green indicate false positive and false negative regions, respectively. Overall, the proposed CTTF produces the cleanest and most structurally consistent change maps, demonstrating superior change perception and boundary regularity across diverse scenarios.

a) WHU-CD (see Fig. 5): The main challenge of this dataset lies in pseudochanges caused by shadows and illumination variations. Most competing methods suffer from boundary-level artifacts near rooftop edges or shaded regions: ChangeFormer, SARASNet, and AFCF3DNet tend to produce overdetection or underdetection along shadow boundaries; Mamba-based models (ChangeMamba and RS-Mamba) alleviate these artifacts to some extent but still show slight

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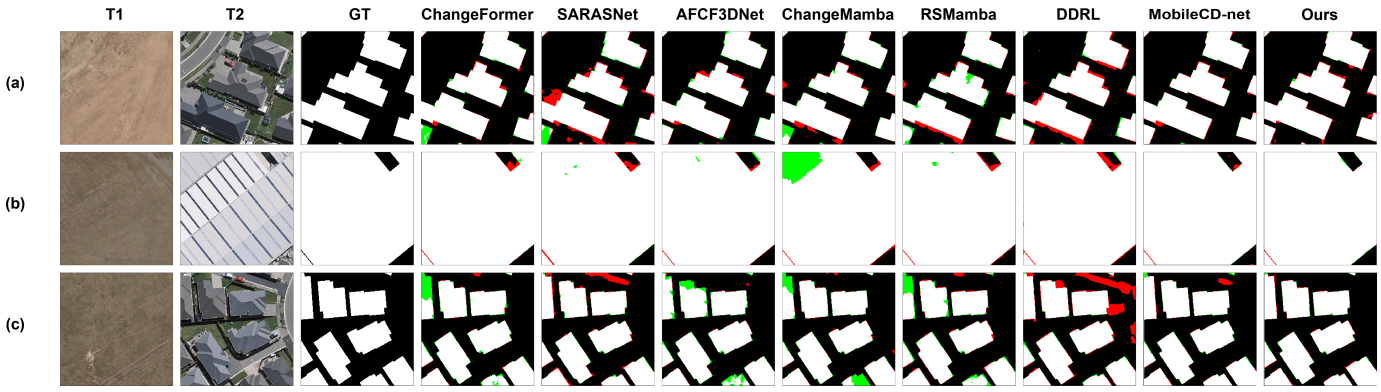

Fig. 5. Visual comparison of CD results on the WHU-CD dataset. (a)–(c) representative samples. White, **black**, **red**, and **green** regions indicate true positive, true negative, false positive, and false negative results, respectively.

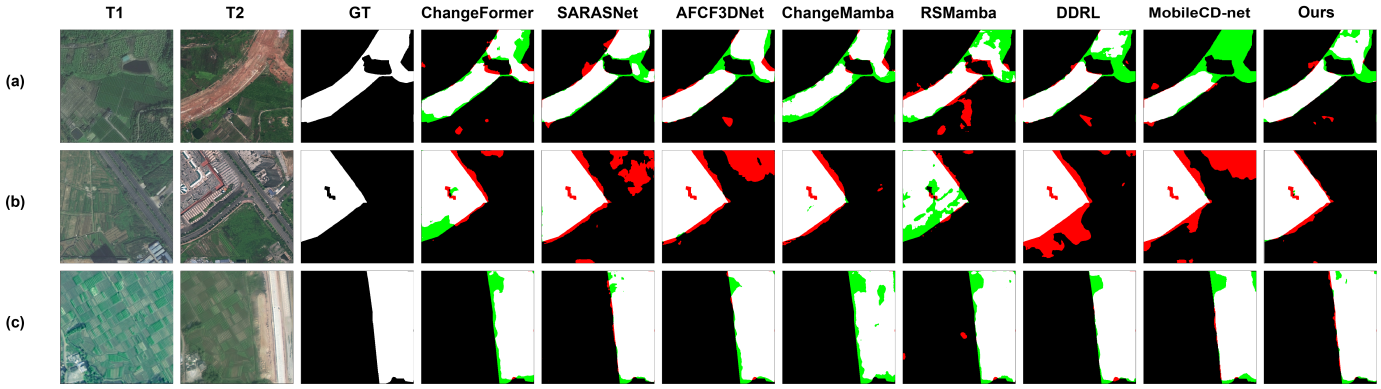

Fig. 6. Visual comparison of CD results on the CLCD dataset. (a)–(c) representative samples. White, **black**, **red**, and **green** regions indicate true positive, true negative, false positive, and false negative results, respectively.

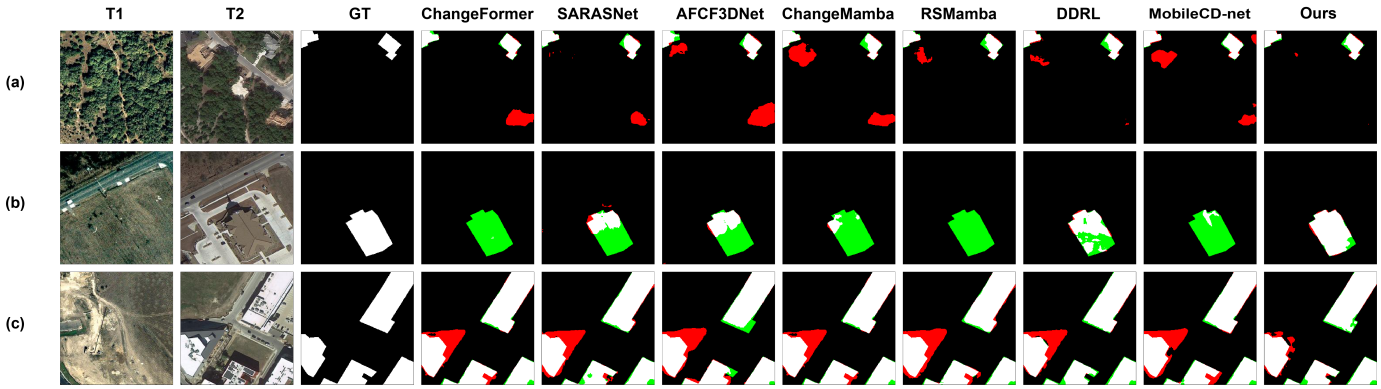

Fig. 7. Visual comparison of CD results on the LEVIR-CD dataset. (a)–(c) representative samples. White, **black**, **red**, and **green** regions indicate true positive, true negative, false positive, and false negative results, respectively.

red/green fringes at corners; and DDRL and Mobile-CDNet introduce fragmented false positives in the background. In contrast, CTTF significantly suppresses such boundary noise, generating smooth and compact building contours well-aligned with the ground truth, without false responses in shaded or unchanged areas.

b) CLCD (see Fig. 6): The CLCD dataset presents greater challenges due to its complex land-cover composition, significant background variations, and vegetation-induced pseudochanges. In case (a), which focuses on road widening scenes, most methods, such as ChangeFormer, SARASNet, and AFCF3DNet, produce obvious overdetections along both sides of the road, while CTTF achieves the most accurate delineation with the smallest detection error. In case (b), representing urban–vegetation mixed regions, other methods generate large red false-positive areas caused by background vegetation, whereas CTTF effectively suppresses such noise, preserving only the true changed regions with regular and clean boundaries. In case (c), covering farmland and bare-ground transitions, models like ChangeMamba, RS-Mamba, and DDRL still produce continuous overdetection belts, while CTTF maintains spatially coherent change regions with smooth and overflow-free edges.

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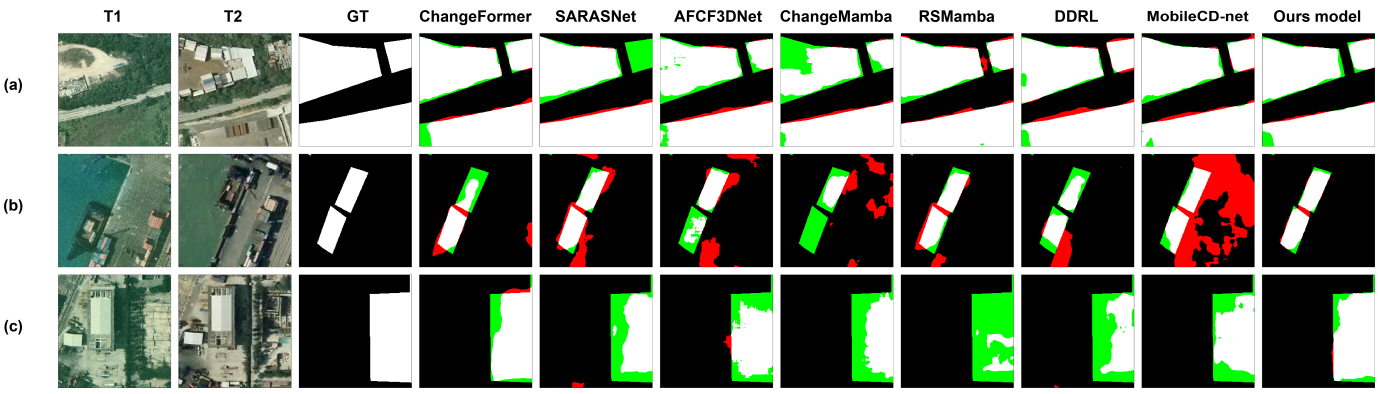

Fig. 8. Visual comparison of CD results on the SYSU dataset. (a)–(c) representative samples. White, **black**, **red**, and **green** regions indicate true positive, true negative, false positive, and false negative results, respectively.

c) Levir-CD (see Fig. 7): This dataset mainly involves cross-temporal building expansions and multiscale structural changes. In case (b), CTTF successfully detects a newly constructed building that is completely missed by ChangeFormer, SARASNet, and AFCF3DNet, while maintaining clean and compact boundaries, revealing its stronger sensitivity to subtle structural differences. In case (c), which contains tilted and small-scale residential buildings, most methods suffer from red/green misalignment strips along oblique or small-object boundaries, whereas CTTF preserves continuous and smooth edges without such artifacts, maintaining stability under rotation and scale variations.

d) SYSU-CD (see Fig. 8): This dataset is more challenging due to complex farmland patterns, irregular object shapes, and severe class imbalance between changed and unchanged regions. In case (a), which involves elongated structures with subtle boundary shifts, most methods, such as ChangeFormer, SARASNet, and AFCF3DNet, exhibit noticeable boundary oscillations, producing red/green artifacts along linear edges, while CTTF maintains smooth and well-aligned contours with minimal deviation. In case (b), representing sparse and small-scale object changes, several methods (e.g., DDRL and Mobile-CDNet) generate excessive false positives in the background, and Mamba-based models still show incomplete detection with fragmented regions. In contrast, CTTF accurately captures the complete change regions while effectively suppressing background noise. In case (c), which contains large homogeneous regions with subtle transitions, competing methods tend to produce oversmoothed or overexpanded predictions, whereas CTTF preserves sharp boundaries and avoids overflow into unchanged areas. Overall, CTTF achieves a favorable balance between completeness and spatial precision, accurately identifying both large-scale and fine-grained changes with minimal noise and distortion.

3) Model Efficiency and Discussion: To further evaluate the efficiency–performance tradeoff, we compare the proposed method with representative approaches on the WHU-CD dataset in terms of model complexity, runtime speed, and detection accuracy. As shown in Table III, lightweight CNN-based methods (e.g., FC-EF and FC-Siam variants) achieve extremely high inference speed with very low computational cost, but their accuracy remains limited. In contrast, Transformer- and Mamba-based methods improve detection performance, while generally incurring higher computational overhead and slower runtime.

TABLE III

COMPARISON OF ACCURACY AND EFFICIENCY. THE BEST RESULTS ARE MARKED IN **BOLD FONT**

Methods	Params	FLOPs	F1	Inference FPS	Train FPS
BIT	11.89	8.71	92.06	283.22	98.31
PaFormer	16.16	10.86	91.85	**346.34**	**103.57**
LCDNet	**2.56**	4.45	90.00	210.39	53.59
AFCF3DNet	17.64	31.58	92.17	151.55	28.68
ACABFNet	102.32	23.93	90.36	102.96	29.18
ChangeFormer	41.03	202.86	90.74	69.20	19.03
RS-Mamba	51.95	22.82	89.72	60.70	18.76
ChangeMamba	48.56	38.49	91.45	47.57	15.00
SNUNet	12.03	46.70	86.95	38.20	12.76
SARASNet	56.89	139.90	83.09	32.38	3.59
Ours	16.21	**3.83**	**94.11**	41.77	15.15

The proposed method achieves the best $F1$ score while maintaining low computational complexity (3.83 GFLOPs) and competitive runtime efficiency. Compared with ChangeFormer and ChangeMamba, our model significantly reduces FLOPs while achieving superior accuracy, indicating a more favorable balance between efficiency and performance. These results suggest that the proposed method is well-suited for practical CD scenarios where both accuracy and efficiency are required.

It is worth noting that, despite the low theoretical complexity, the training speed of the proposed method is not the fastest among all compared approaches. This is mainly attributed to the architectural characteristics of the Mamba-based modules. In particular, the selective scanning operation introduces inherent sequential dependencies, which constrain parallel execution on current GPU implementations. Consequently, the practical runtime does not fully reflect the theoretical computational advantage.

In addition, the current implementation has not yet been fully optimized for hardware acceleration. With further engineering improvements and kernel-level optimization, the runtime efficiency can be further enhanced. Overall, the pro-

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TABLE IV
ABLATION STUDY OF THE PROPOSED MODULES ON WHU-CD. THE BEST RESULTS ARE MARKED IN **BOLD FONT**

Model	OA	Pre.	Rec.	F1	IoU
Baseline	99.26	90.81	93.27	92.02	85.22
+SASF	99.31	94.05	90.78	92.38	85.85
+GLP	99.37	95.94	90.05	92.90	86.74
+GLP (w/o PA) + SASF	99.33	95.76	89.32	92.42	85.92
+GLP + SASF (Full)	**99.47**	**95.62**	**92.65**	**94.11**	**88.88**

posed method still provides a favorable tradeoff between detection performance and computational cost.

D. Ablation Studies

To comprehensively evaluate the contribution of each proposed component, a series of ablation experiments was conducted on the WHU-CD dataset under the same training and inference configurations as the baseline. The proposed decoder-side framework CTTF introduces two modules: the GLP Block for boundary-aware GLP decoding and the SASF Module for explicit tri-difference integration. Table IV summarizes the quantitative results in terms of OA, precision, recall, $F1$, and IoU.

1) Effectiveness of SASF: Introducing the SASF Module significantly improves the $F1$ from 92.02% to 92.38% and the IoU from 85.22% to 85.85%. By explicitly constructing the signed, absolute, and sum differences and using the sum branch as a stability prior, the SASF Module performs multiscale calibration between difference types, thereby suppressing pseudochanges and recovering weak variations that are easily missed under misregistration.

2) Effectiveness of GLP: Substituting the SASF Module with the GLP Block further increases the $F1$ to 92.90% and the IoU to 86.74%. GLP disentangles global semantics, local high-frequency details, and pointwise anchors through the combination of Mamba-based selective scanning, wavelet-based multiresolution decomposition, and multiscale convolution. This structure achieves a marked precision gain, evidencing sharper boundaries and reduced edge-region false alarms.

3) Role of PA: When PA is removed from the GLP branch, the model performance drops slightly ($F1$: 92.42% and IoU: 85.92%). This confirms that PA acts as a crucial pointwise anchoring mechanism that stabilizes subtle edge responses and preserves tiny structures during decoding.

4) Complementarity of GLP and SASF: Activating both the GLP Block and the SASF Module with PA yields the best results, reaching an $F1$ of 94.11% and an IoU of 88.88%, corresponding to a relative gain of +2.09% and +3.66% over the baseline. The joint framework simultaneously preserves global semantic consistency, refines boundary precision, and regularizes difference cues through a stability prior, effectively overcoming the decoder-side limitations of pure global modeling and single-path difference bias.

5) Discussion of Full Structure: As shown in Fig. 9, both GLP and SASF bring clear improvements over the baseline. Each module effectively enhances structural completeness and suppresses pseudochanges, leading to fewer fragmented boundaries and fewer background false alarms. The performance gap between using GLP alone and SASF alone is relatively small, and both yield consistently stable results across different scenes.

When applying GLP (without PA) + SASF, a distinct tradeoff can be observed: false positives slightly increase, while false negatives are noticeably reduced. This indicates that the PA branch still plays an important role in fine-grained noise suppression and pixel-level refinement. Overall, the full GLP + SASF configuration achieves the best outcomes. It provides the most accurate contours, the cleanest background, and the closest match to the ground truth across all examples. These results highlight the strong complementarity between global–local structural modeling (GLP) and multiperspective difference fusion (SASF), demonstrating their combined effectiveness in improving visual quality and robustness.

The progressive improvements demonstrate that decoder-side design is pivotal for Mamba-based CD. The SASF Module enhances difference completeness and robustness by integrating complementary tri-difference information, while GLP strengthens boundary precision through a balanced GLP representation. Their integration in CTTF achieves fine-grained yet stable change localization under illumination variations and minor misregistration, attaining state-of-the-art (SOTA) performance with minimal computational overhead.

6) Sensitivity Analysis on GLP Weight Ratios: To determine the optimal balance among the four subbranches in the GLP Block, we conducted a sensitivity study on their relative ratios. The four branches correspond to the global path, local path, point path, and identity path. Since the GLP units in the last two decoder stages have a relatively minor impact on overall performance, we fixed their ratios as global-ratio = 0.25, local-ratio = 0.25, pa-ratio = 0, and identity-ratio = 0.5. The parameter adjustment was only applied to the GLP modules in the first two stages of the decoder.

As shown in Table V, excessively high or low weights assigned to any single branch lead to degraded performance. When the global and local ratios are both fixed at 0.25 and the point ratio gradually increases from 0 to 0.25, the $F1$ first improves and then drops. A moderate inclusion of the PA path (pa-ratio = 0.2) yields the best performance ($F1$ = **94.11%** and IoU = **88.88%**), whereas higher ratios (> 0.3) overemphasize localized activations, compromising global consistency and slightly increasing false positives.

The global and local paths play complementary roles: the global branch ensures structural coherence across the scene, while the local branch enhances boundary details and fine textures. Balanced proportions (around 0.2–0.25 for each) allow the decoder to capture both long-range dependencies and local variations effectively. In contrast, when the identity path dominates (e.g., identity-ratio = 0.5), the decoder becomes less adaptive, causing a small drop in $F1$ and IoU.

In summary, the sensitivity analysis indicates that the optimal GLP configuration is global-ratio = 0.2, local-ratio = 0.2, pa-ratio = 0.2, and identity-ratio = 0.4. This balanced setting provides the best tradeoff between semantic stability

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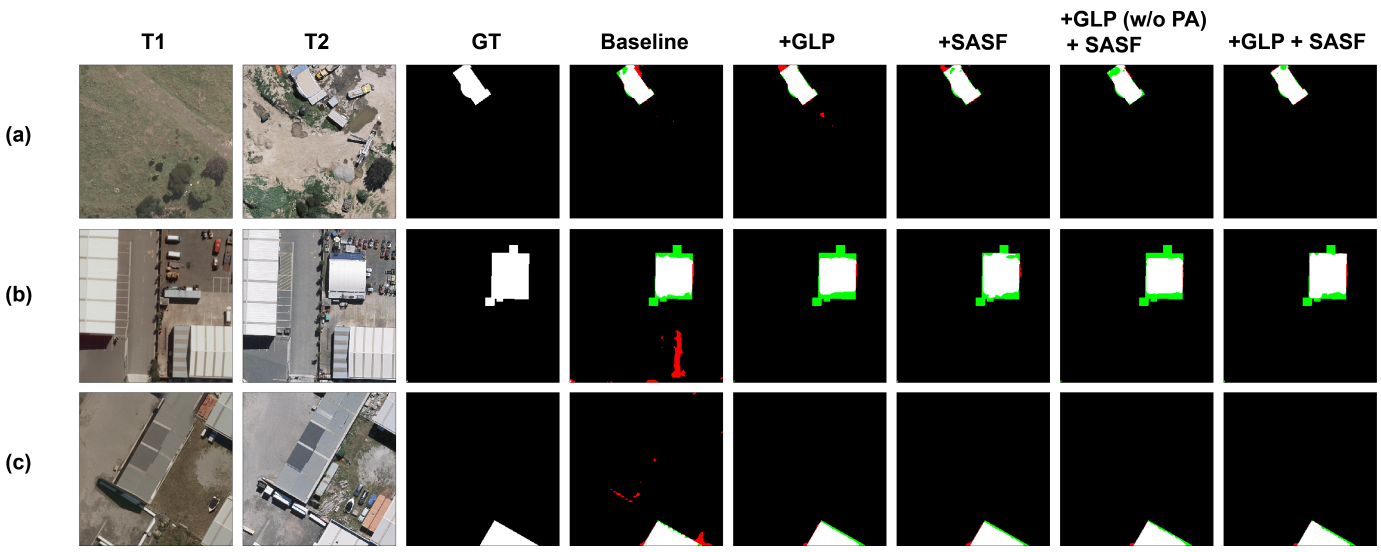

Fig. 9. Visual comparison of CD results on the WHU dataset. (a)–(c) representative samples. White, **black**, **red**, and **green** regions indicate true positive, true negative, false positive, and false negative results, respectively.

TABLE V

DECODER WEIGHT CONFIGURATIONS (CFG-1–CFG-8). EACH 2×2 MATRIX DENOTES GLOBAL ("G"), LOCAL ("L"), POINT ("P"), AND IDENTITY ("I")

		Decoder Stages (Cfg-1 to Cfg-8)							
item	**note**	**Cfg-1**	**Cfg-2**	**Cfg-3**	**Cfg-4**	**Cfg-5**	**Cfg-6**	**Cfg-7**	**Cfg-8**
weight matrix	$\begin{bmatrix} G & L \\ P & I \end{bmatrix}$	$\begin{bmatrix} 0.25 & 0.25 \\ 0.00 & 0.50 \end{bmatrix}$	$\begin{bmatrix} 0.25 & 0.25 \\ 0.10 & 0.40 \end{bmatrix}$	$\begin{bmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{bmatrix}$	$\begin{bmatrix} 0.20 & 0.20 \\ 0.20 & 0.40 \end{bmatrix}$	$\begin{bmatrix} 0.20 & 0.20 \\ 0.30 & 0.30 \end{bmatrix}$	$\begin{bmatrix} 0.15 & 0.15 \\ 0.40 & 0.30 \end{bmatrix}$	$\begin{bmatrix} 0.15 & 0.15 \\ 0.50 & 0.20 \end{bmatrix}$	$\begin{bmatrix} 0.10 & 0.10 \\ 0.60 & 0.20 \end{bmatrix}$
Pre.	(%)	95.14	95.35	96.07	95.62	92.94	94.58	95.09	94.94
Rec.	(%)	91.38	92.08	89.83	92.65	92.87	92.46	91.19	91.37
F1	(%)	93.22	93.69	92.85	94.11	92.91	93.51	93.10	93.12
IoU	(%)	87.30	88.13	86.65	88.88	86.75	87.81	87.09	87.13
OA	(%)	99.39	99.40	99.35	99.47	99.35	99.41	99.38	99.38

and boundary refinement, confirming that the GLP synergy is essential for achieving high-precision CD within the CTTF framework.

7) Further Investigation on SASF: To further verify the effectiveness of the proposed tri-difference mechanism, we performed an additional ablation on the three constituent branches of the SASF Module, namely the signed difference ($-$), the absolute difference ($|-|$), and the sum difference ($+$). Each variant retains one or several difference paths to examine their individual and joint contributions.

As shown in Table VI, using a single difference type (signed, absolute, or sum) leads to limited performance, with $F1$ ranging from 92.46% to 92.98%. Among these, the absolute difference achieves slightly higher stability by focusing on magnitude cues but lacks directional sensitivity. The signed difference provides complementary polarity information yet suffers from amplitude bias and noise sensitivity, while the sum difference alone offers better global consistency but insufficient discriminative strength for local variations.

When combining two difference forms, performance improves across all metrics. The signed+sum configuration yields a significant leap in recall (92.65%) and IoU (87.65%), suggesting that the sum branch effectively compensates for the sign branch's instability and strengthens the model's robustness under slight misregistration. The absolute+sum variant also improves recall and $F1$ but to a lesser extent, indicating that the signed branch contributes more to directional contrast than the absolute branch alone. The best performance is obtained when all three difference forms are activated

TABLE VI

ABLATION ON INDIVIDUAL AND COMBINED BRANCHES OF SASF MODULE. "$-$," "$|-|$," AND "$+$" ARE SIGNED DIFFERENCE, ABSOLUTE DIFFERENCE, AND SUM DIFFERENCE, RESPECTIVELY. THE BEST RESULTS ARE IN **BOLD FONT**

$-$	$\lvert-\rvert$	$+$	**Pre.**	**Rec.**	**F1**	**IoU**
✓	×	×	94.24	90.76	92.46	85.99
×	✓	×	94.37	91.64	92.98	86.89
×	×	✓	94.51	90.50	92.51	86.05
✓	✓	×	94.52	91.11	92.79	86.54
✓	×	✓	94.21	92.65	93.42	87.65
×	✓	✓	93.79	91.90	92.84	86.64
✓	✓	✓	**95.62**	**92.65**	**94.11**	**88.88**

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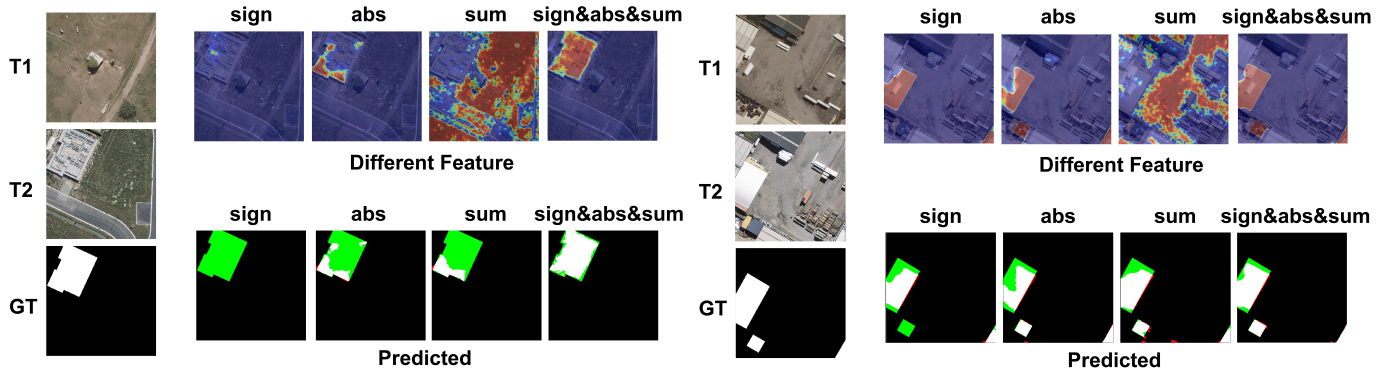

Fig. 10. Visual comparison of CD predicted results and differential feature maps on the WHU dataset. "sign," "abs," and "sum" denote signed difference, absolute difference, and sum difference, respectively. In the different feature section, the red region indicates higher attention values, while the blue region indicates lower attention values. In the predicted section, white, **black**, **red**, and **green** regions indicate true positive, true negative, false positive, and false negative results, respectively.

simultaneously (signed+absolute+sum), achieving a $F1$ of **94.11%** and IoU of **88.88%**.

Fig. 10 presents a comparison of differential feature maps generated by three individual branches—signed, absolute, and sum—as well as their joint fusion (signed & absolute & sum), together with the corresponding predicted results. For both examples, the signed branch captures only coarse outline-level differences and misses many subtle changed areas, leading to noticeable false negatives in the predicted maps. The absolute branch highlights pixel-level intensity discrepancies but is susceptible to background illumination variations, resulting in several false positives. The sum branch responds to large structural differences, yet it also amplifies irrelevant spectral fluctuations, producing noisy, cluttered activation maps.

In contrast, the fused signed & absolute & sum features exhibit the most consistent and complete activation patterns: true changed regions are clearly emphasized, while irrelevant background variations are well suppressed. This leads to the most accurate predicted masks, with fewer fragmented errors compared to using any single branch alone. Fig. 10 demonstrates that no individual difference representation is sufficient, and meaningful change cues are distributed across multiple differential domains. The complementary integration of signed, absolute, and sum differences provides a more reliable and discriminative change representation, directly contributing to improved prediction quality. This configuration integrates the directional sensitivity of the signed difference, the magnitude robustness of the absolute difference, and the stability prior of the sum difference. The complementarity among these three cues allows the SASF Module to explicitly capture both structural and radiometric variations while maintaining temporal consistency.

In summary, this fine-grained ablation validates the design rationale of the tri-difference fusion mechanism. By balancing polarity, magnitude, and stability priors, the SASF Module achieves more reliable change localization, particularly in regions affected by illumination variations, subtle texture changes, or registration noise.

8) Sensitivity Analysis on SASF Weight Ratios: To further explore the effect of branch weighting within the SASF Module, a sensitivity analysis was conducted on the relative proportions of the local feature extraction branch and the identity branch. In this experiment, the parameter "Scale" denotes the proportion assigned to the local branch, while the identity branch automatically occupies the remaining weight. For example, a scale value of 0.1 indicates that the local path accounts for 10% of the total weight and the identity path occupies the remaining 90%. The quantitative results are shown in Table VII.

TABLE VII
MULTISCALE EVALUATION OF THE SASF MODULE. THE BEST RESULTS ARE IN **BOLD FONT**

Scale	OA	Pre.	Rec.	F1	IoU
0	99.40	94.78	92.05	93.39	87.61
0.1	**99.47**	95.62	92.65	**94.11**	**88.88**
0.2	99.41	95.14	91.83	93.46	87.72
0.3	99.41	94.27	92.79	93.53	87.84
0.4	99.35	94.97	90.71	92.79	86.55
0.5	99.38	93.38	**93.06**	93.22	87.30
0.6	99.42	94.58	92.78	93.67	88.10
0.7	99.37	95.18	90.79	92.93	86.80
0.8	99.34	95.04	90.36	92.64	86.28
0.9	99.40	95.48	91.20	93.29	87.42
1.0	99.43	**95.81**	91.48	93.60	87.96

From the results, a clear performance trend can be observed. When the local ratio increases from 0 to 0.1, both the $F1$ and IoU improve significantly, from 93.39% to 94.11% and from 87.61% to 88.88%, respectively. This demonstrates that introducing a small proportion of local multiscale feature extraction enhances fine-grained detail representation and boundary refinement. The additional local processing helps to capture subtle variations and texture discontinuities that are easily overlooked when relying solely on global or identity propagation. However, as the local ratio continues to rise beyond 0.2, the model performance begins to decline gradually. Excessive local weighting introduces redundant neighborhood information and disrupts the stability provided by the identity branch, resulting in weaker global consistency and slightly increased false-positive rates around complex

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TABLE VIII
ABLATION ON THE DIFFERENTIAL REFINEMENT STRATEGY IN THE SASF MODULE. THE BEST RESULTS ARE MARKED IN **BOLD FONT**

Refinement Strategy	OA	Pre.	Rec.	F1	IoU
No refinement	99.39	95.08	91.54	93.28	87.41
Global-only	99.34	94.11	91.29	92.68	86.35
Global + Local	99.26	93.22	90.48	91.83	84.89
Local + PA	99.30	91.97	**92.82**	92.40	85.87
Local-only (Ours)	**99.47**	**95.62**	92.65	**94.11**	**88.88**

edges. When the local ratio exceeds 0.5, $F1$ fluctuates between 92.6% and 93.6%, confirming that overemphasizing local cues diminishes the discriminative power of the decoder. The configuration with a local ratio of 0.1 yields the most balanced result, achieving the highest $F1$ (94.11%) and IoU (88.88%) with a high OA of 99.47%. This indicates that a small but nonnegligible contribution from the local branch effectively complements the global contextual representation while retaining the structural stability offered by the identity connection. The identity path acts as a residual constraint that preserves temporal consistency and prevents overfitting to local noise, while the local branch adaptively enriches detailed features.

In summary, the sensitivity study validates that the SASF Module benefits from a lightweight local enhancement (10%) coupled with a dominant identity path (90%), which together provide the optimal tradeoff between detail awareness and semantic stability. This balanced configuration ensures more precise boundary delineation and improved robustness, further reinforcing the overall effectiveness of the CTTF framework.

9) Ablation on Differential Refinement Strategy in SASF: To further investigate the design choice of the refinement mechanism in the SASF Module, we conduct a series of ablation experiments by varying the refinement strategies applied after the tri-difference construction. Specifically, we consider five configurations: 1) removing the refinement module entirely (no refinement); 2) using only the global branch (global-only); 3) combining global and local branches (global+local); 4) introducing PA into the local branch (local+PA); and 5) the proposed setting that retains only a lightweight local branch (local-only).

As shown in Table VIII, the proposed local-only refinement achieves the best performance across all metrics, with precision, recall, $F1$ score, and IoU reaching 95.62%, 92.65%, 94.11%, and 88.88%, respectively. This result demonstrates that restricting the refinement to a lightweight local branch is more effective in enhancing intertemporal structural discrepancies, particularly for boundary localization and fine-grained change representation.

The no-refinement variant also achieves competitive performance, indicating that the tri-difference representation itself already provides strong discriminative capability. However, its $F1$ score and IoU (93.28% and 87.41%) are consistently lower than those of the local-only configuration, suggesting that additional local refinement helps further improve boundary completeness and structural consistency.

In contrast, both global-only and global+local configurations lead to noticeable performance degradation. In particular, the global+local setting reduces the $F1$ score and IoU to 91.83% and 84.89%, respectively. This observation indicates that reintroducing global modeling in the differential space is not only redundant but also detrimental. Since global semantic information has already been sufficiently captured by the encoder and GLP Block, applying additional global aggregation at the differential stage tends to smooth out local discrepancies, thereby weakening boundary sharpness and changing localization accuracy.

Furthermore, the local+PA configuration achieves a slightly higher recall (92.82%) but suffers from reduced precision, resulting in a lower overall $F1$ score and IoU compared to the local-only setting. This suggests that while PA can enhance sensitivity to certain subtle changes, it may also amplify noise in the differential domain, leading to less stable predictions.

Overall, these results validate the core design principle of SASF: in the differential space, lightweight local refinement is sufficient and preferable, whereas additional global modeling is unnecessary and may even degrade performance in our setting. By focusing on localized discrepancy correction rather than redundant semantic aggregation, the proposed design achieves superior performance in CD.

V. CONCLUSION

This article systematically proposes a novel decoder architecture for remote sensing CD, namely CTTF. Structurally, CTTF introduces a GLP Block with four decoupled branches, including global, local, PA, and identity branches, for multiscale feature reconstruction and boundary refinement. The global branch couples Mamba's selective scanning with wavelet transform, the local branch adopts multireceptive-field dynamic convolutions, the PA branch enhances subtle changes, and the identity branch stabilizes feature propagation. This design improves boundary discrimination and structural fidelity while maintaining semantic consistency.

From the differential perspective, CTTF proposes the SASF Module, which unifies signed, absolute, and sum differences. With multiscale convolutions and channelwise fusion, SASF strengthens directional, magnitude, and contextual cues, suppresses pseudochanges, and improves sensitivity to weak and fine-grained variations.

Extensive experiments demonstrate that CTTF achieves SOTA performance on four datasets. Ablation studies further confirm the complementary contributions of the GLP Block and SASF Module to boundary precision, differential completeness, and model robustness.

In summary, CTTF enhances local sensitivity and cross-temporal differential expressiveness while preserving global consistency, offering a fine-grained and generalizable decoder design for high-resolution CD. Nevertheless, several aspects remain worthy of further investigation. First, although the current design enhances the interaction between global and local features in the decoding stage, further optimization toward more lightweight and efficient architectures may improve scalability when handling ultrahigh-resolution remote sensing imagery. Second, more adaptive mechanisms could be

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explored to dynamically adjust the contribution of different differential pathways under diverse imaging conditions, which is also consistent with recent studies that emphasize adaptive aggregation and structural modeling in remote sensing analysis [30], [31]. Third, the proposed framework is mainly evaluated on optical remote sensing datasets. Extending it to more complex scenarios, such as multimodal CD, SAR-optical data analysis, and hyperspectral imagery, may further improve its robustness and applicability in diverse environments. Finally, inspired by broader developments in related visual and remote sensing tasks [17], [18], [19], [20], [21], future work will also explore the transferability of CTTF to more diverse dense prediction tasks, self-supervised settings, and large-scale dynamic monitoring.

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